

Last Time:

□ Experimental Design

This Time:

□ collab. / coord. / Assist.

lecture 9

HRI, FALL '26

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Collaboration / Assistance / Coordination

- **COLLABORATION**: ex. R and H are moving a couch together
 - R & H share the same objective
 - —"—— share the workspace (w/ two diff. embodiments)
 - —"—— both often know the objective
 - —"—— act as a team $\Rightarrow$ R must plan jointly w/ H
- **ASSISTANCE**: ex. robot is being teleoperated by human to skewer food
 - R & H share the same objective
 - —"—— share the workspace (via the shared embodiment)
 - only H perfectly knows the objective
 - R's job is to help, despite uncertainty $\Rightarrow$ R must infer & then assist
 - $\hookrightarrow$ preserve some autonomy of H!
- **COORDINATION**: ex. autonomous car & human car passing @ intersection.
 - R and H have different but not necessarily opposing objectives
(ex. R $\rightarrow$ left lane w/o crash
H $\rightarrow$ right lane w/o crash)
differ share
 - —"—— share the workspace (w/ diff. embodiments)
 - H's objective is not necessarily known to R & vice versa
 - R must coordinate w/ H to accomplish its goal w/o hindering H.

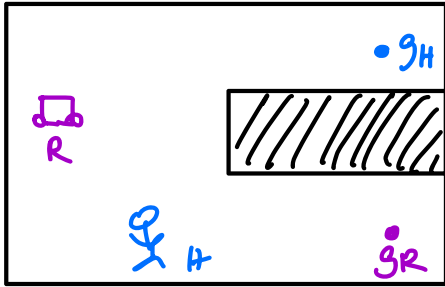
Q How do we model these diff. types of interactions mathematically?

⇒ TODAY: MDPs / POMDPs

⇒ NEXT WEEK: Game Theory 😊

INTERACTION AS AN MDP / POMDP

• COORDINATION - Navigation example



→ model the robot's behavior in a shared workspace w/ human, whose behavior we (the robot) do NOT control

→ R must avoid H while also being efficient in getting to gR

• state: $s = (s_H, s_R)$ // joint state space of H & R

• action: $a := a_R \in \mathcal{A}_R$ // robot only controls its own actions!
(this MDP is from the "robot's POV")

• reward:

$$r(s, a) = \begin{cases} -100 \cdot \mathbb{1}\{s_H = s_R\} & // \text{safety wrt. H} \\ -10 \cdot \mathbb{1}\{s_R \in \text{obs}\} & // \text{safety wrt. env.} \\ -\text{dist}(s_R, g_R) & // \text{efficiency} \\ \vdots \end{cases}$$

• transition:

by defn

$$P(s' | s, a) = P(s'_R, s'_H | s_R, s_H, a_R)$$

product rule

$$= P(s'_R | s'_H, s_R, s_H, a_R) \cdot P(s'_H | s_R, s_H, a_R)$$

Robot dynamics ✓

$= P(s'_R | s_R, a_R)$ b/c given R's current state & action, R's next state is independent of H. ⊗

$$\textcircled{*} P(s_H' | s_R, s_H, a_R) = \sum_{a_H} P(s_H', a_H | s_H, s_R, a_R) \quad // \text{marginalize out } H's \text{ control action}_1$$

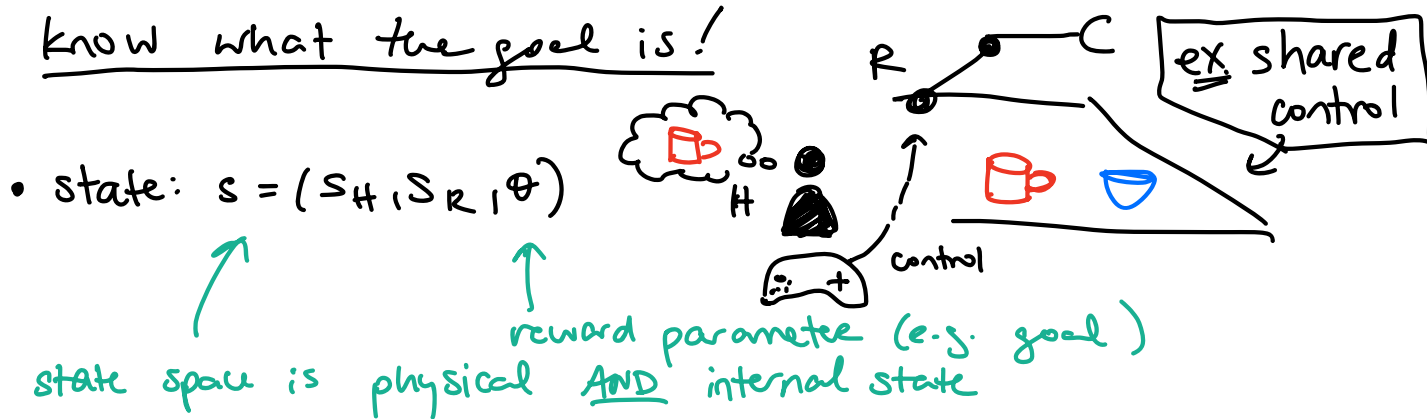
$$= \sum_{a_H} \underbrace{P(s_H' | s_H, s_R, a_R, a_H)}_{\downarrow} \cdot P(a_H | s_H, s_R, a_R) \quad // \text{prod. rule}$$

$P(s_H' | s_H, a_H)$ b/c given H 's state + action the H 's next state is indep. of R .

$$= \sum_{a_H} \underbrace{P(s_H' | s_H, a_H)}_{H's \text{ dynamics / transition func.}} \cdot \underbrace{P(a_H | s_H, s_R)}_{H's \text{ policy (predictive model!)}} \quad \text{with } (s_H, s_R) \text{ above the second term}$$

• ASSISTANCE as a POMDP

So far, robot had its own known goal/reward; now the R is helping H achieve their goal, BUT it doesn't know what the goal is!



- $a := a_R \in A_R$ // robot only controls itself
- $r(s, a) = \text{"achieve } \theta"$ // reward means executing the task the way the H wants
- $o := a_H \in A_H$ // observations of human's actions! use this as evidence for inferring θ .
- transition: $P(s' | s, a) = (\text{next page})$

$$= P(s_H', s_R' | \theta' | s_R, s_H, \theta, a_R)$$

common assumption that $\theta' = \theta$ but not true in general

⋮ (exercise 😊)

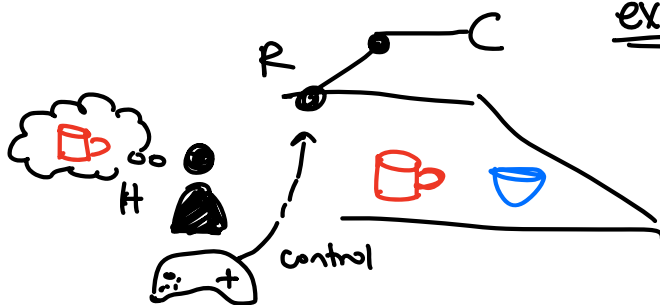
$$= \underbrace{P(s_R' | s_R, a_R)}_{R \text{ dynamics}} \cdot \sum_{a_H} \underbrace{P(s_H' | s_H, a_H)}_{H \text{ dynamics}} \cdot \underbrace{P(a_H | s_H, \theta)}_{H's \text{ policy } \boxed{\text{given intent}}}$$

• observation model:

$$P(o | s) = P(a_H | s_H, s_R, \theta) \stackrel{\text{assump.}}{=} P(a_H | s_H, \theta) \quad \left. \vphantom{P(a_H | s_H, \theta)} \right\} \begin{array}{l} \text{given by} \\ H \text{ and} \\ \text{their policy} \end{array}$$

IN THEORY, solving this for the robot's optimal assistance strategy would involve the robot maintaining a belief $b(\theta) := P(\theta | a_H^{0:t}, s_H^{0:t})$ and solving POMDP to yield information-gathering actions!

IN PRACTICE, very hard to do but, a common approx. is to solve something called a Q-MDP



ex. SHARED CONTROL

① infer which goal H wants

$$b^0(\theta) = \begin{array}{c} \text{red bar} \quad \text{blue bar} \end{array}$$

$$\vdots$$

$$b^t(\theta) = \begin{array}{c} \text{red bar} \quad \text{blue bar} \end{array}$$

② robot policy via QMDP:

$$a_R^* = \arg \max_{a_R} \sum_{\theta} b(\theta) \cdot Q_{\theta}(s, a_R)$$

③ blend human's command with robot's assistive action

$$a_{\text{exec.}} = (1-\alpha) a_H + \alpha a_R^*, \quad \alpha \in [0,1]$$

if you want to see more blending schemes, see

[Dragon & Srinivasa, 2013]!

→ we will read this paper in class too!

④ What are some challenges you foresee with this shared controller?

→ e.g. is a_H (low level H action) "enough" to understand intent in full complexity of the world?

→ e.g. joystick is low-Dof but that then gets translated to high-Dof robot!

See:
[Jeon, Losey, Sadigh.
"Shared Autonomy with
Learned Latent Actions"
RSS 2020.]

u_H 